

The Effect of AI Tools on Developer Wages by Experience Level

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Abstract

This paper studies whether AI tools affect developer wages differently depending on their level of professional work experience. Using the 2025 Stack Overflow Annual Developer Survey, I estimate a log-wage Ordinary Least Squares regression with an interaction term between AI use and years of experience, controlling for education, firm size, industry, and country. Contrary to prior productivity literature on AI, which find that less-experienced workers benefit most from using AI, I find that AI tools complement rather than substitute for experience: every additional year of experience roughly pays 2.0% for non-AI users and 3.3% for AI users, with the benefit of using AI being concentrated among senior developers. The study highlights how the productivity effect apparent in experimental and controlled settings does not appear to translate onto actual developer salaries.

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Introduction

Over these last few years, AI tools have changed from a curiosity to an everyday part of how developers write code. GitHub Copilot and ChatGPT have changed what productivity can look like. An unprecedented amount of software work is increasingly being offloaded to a tool that sits one keystroke away, with major firms flaunting how much of their codebases is now AI-generated. As a computer science and economics double major planning for data-related roles, including possible programming after graduation, I want to know whether learning these tools will pay off in the labor market and whether it looks different for someone just starting their career vs. someone with many years of experience. That's the question this paper answers: do AI tools affect wages differently depending on a developer's experience?

Acemoglu and Restrepo (2018) provide the theoretical groundwork. Although they frame their research in AI and productivity instead of wages, they argue that AI can either substitute labor by performing routine tasks that workers used to perform, lowering wages, or complement long-time workers by raising their productivity in complex, nuanced tasks, raising wages. However, theory cannot predict which force dominates; it must be tested empirically. Therefore, I take a look at recent empirical studies, all of which point the force in one direction: AI substitutes for experience, raising productivity for the least experienced workers. What none of these studies do, though, is look directly at wages; they all stop at productivity. My paper picks up where they leave off and asks whether the productivity that AI brings to the table actually translates into pay.

To answer this question, I run a cross-sectional, log-linear regression on the 2025 Stack Overflow Annual Developer Survey, a global survey of more than 49,000 developers across 177 countries that voluntarily collects information on the state of software development such as but not limited to developer salary, professional experience, AI tool use, education, firm size, industry, and country, all factors that I take into account in my model. More specifically, my dependent variable is log annual salary, while my independent variables include a dummy for AI use, years of professional work experience, an AI $\times$ Experience interaction term, and four binary controls for education, firm size, industry, and country. The coefficient of interest is the interaction term, as it tells me whether each additional year of experience pays off less or more for developers who use AI than for those who don't. If AI substitutes for experience the way the productivity literature suggests, the interaction coefficient should be negative.

To my surprise, it isn't! The interaction coefficient is statistically significant, but in the opposite direction I originally hypothesized. AI tools seem to complement experience rather than substitute for it: every additional year of experience is worth about 65% more in wages for AI users than for non-AI users (3.3% vs. 2.0% per year). The total AI effect on wages also varies depending on experience. Below 11.38 years of experience, AI users earn less than non-AI users, and above that, AI users earn more. The benefit of AI use largely goes to senior developers, contrary to what the productivity literature indicates. Even after correcting for heteroskedasticity or for functional form, this result persists.

Literature Review

The question of how technology shapes wages is one of the oldest in labor economics, and the fact that it does so differently across workers makes the AI literature particularly intriguing. The theoretical backbone of my paper starts with Acemoglu and Restrepo (2018), who argue that automation technologies like AI can substitute labor by performing routine tasks that workers used to perform, reducing employment and wages, or they can complement long-time workers by raising their productivity toward complex tasks, increasing employment and wages. Theory can not predict which force dominates, as it depends on the technology, the task at hand, and the worker, meaning it has to be answered empirically. That framing is essentially the motivation for my interaction term: I am asking which of the two forces dominates for software developers, and whether the answer depends on their level of professional work experience.

A useful way to think about why experience matters comes from human capital theory, in which wages reflect the skills, knowledge, and on-the-job training that workers acquire. If AI tools can replicate the knowledge that experienced developers have built up over the years, such as knowledge of design patterns and best debugging practices, then AI substitutes for experience; the pay gap between junior and senior developers should shrink. But if AI amplifies what experienced developers already do well, then AI complements their experience; the pay gap should widen. The coefficient of interest, β_3 , is essentially a test of which of these two stories the data actually support.

The recent empirical literature all points to one of these stories. Brynjolfsson, Li, and Raymond (2025), in a study of more than 5,000 customer service agents working for an undisclosed Fortune 500 firm, find a 14% average productivity gain from access to a generative AI assistant, with 34% for novice workers and close to zero for experienced ones. Their interpretation is that the AI tool captures the ‘common’ knowledge that senior workers had accumulated and passes it along to less experienced workers, closing the productivity gap between them. Peng et al. (2023) find a similar pattern in software development: in a randomized controlled trial with 95 professional developers, those given access to GitHub Copilot completed a coding task 55.8% faster than the control group, with the largest gains again being documented among less experienced developers. Cui et al. (2024) extend this to a much larger field experiment at Microsoft and Accenture covering nearly 2,000 developers, where AI-assisted developers produced 13-22% more pull requests per week, with junior developers benefiting most. These studies done across different industries and circumstances all point to the same consistent pattern: AI substitutes for experience, raising productivity for the least experienced workers.

What this prior research has not yet established, and what my paper is trying to figure out, is whether this productivity actually shows up in wages. The studies above all measure output: chats resolved, tasks completed, pull requests merged, rather than pay. However, a worker can become more productive without their wage moving immediately, especially in firms with sticky wages or where AI use does not yet compensate workers in the labor market. That is why my regression estimates whether wages actually reflect AI’s productivity effects and if they differ by experience.

There's also a related body of work showing that wages vary a lot depending on the size of the firm someone works for, patterns that Brown and Medoff (1989) document, and the country they work in, findings that Freeman and Oostendorp (2000) note. That's why I include controls for company size, industry, and country in my analysis.

Taken together, the prior literature gives me an equally clear theoretical and empirical prediction. Theory says AI can either substitute for or complement experience, and the productivity evidence suggests substitution as the force that dominates; AI raises the output of less-experienced workers more than that of experienced ones. If wages reflect productivity, then my interaction term β_3 should be negative, meaning each additional year of experience pays off less for developers who use AI than for those who don't. AI would narrow the wage gap between junior and senior developers the same way it narrowed the productivity gap in customer service and coding studies.

Model

Project Map

The project map pictured below (see Figure 1) showcases several important determinants of a developer's annual salary. The variables that I ultimately included in my main regression are in blue, while some excluded factors that the model doesn't directly measure are likely sources of omitted variable bias (OVB), which I revisit in the Results section.

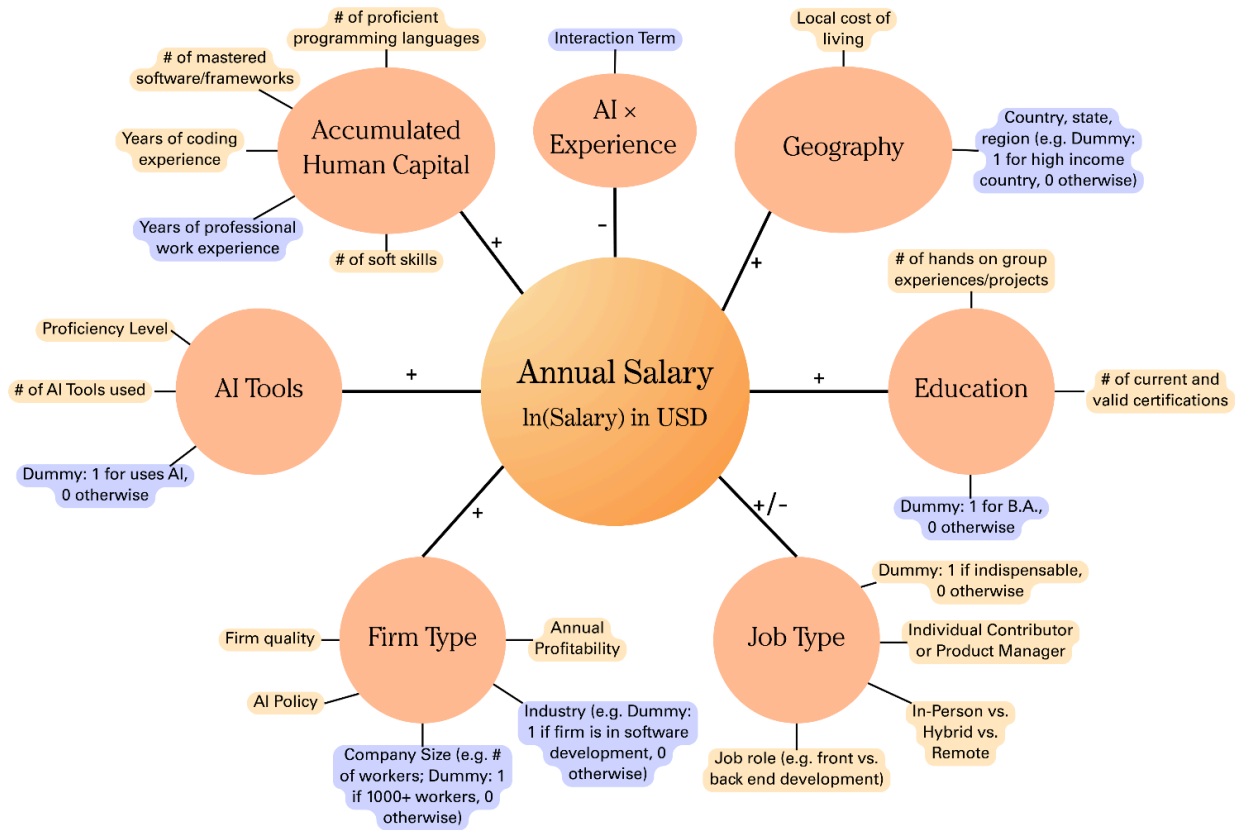


Figure 1. Project map for log developer salary, distinguishing between measured and excluded variables.

Specification

$$\ln(\text{Salary}_i) = \beta_0 + \beta_1 \text{AI}_i + \beta_2 \text{Exp}_i + \beta_3 (\text{AI}_i \times \text{Exp}_i) + \beta_4 \text{Edu}_i + \beta_5 \text{Size}_i + \beta_6 \text{Industry}_i + \beta_7 \text{Country}_i + \varepsilon_i$$

The dependent variable, $\ln(\text{Salary}_i)$, is the natural log of developer i 's 2025 annual salary measured in U.S. dollars. The independent variables are: AI_i , a dummy variable equal to 1 if the i th developer reports uses AI tools in their workflow; Exp_i , years of professional work experience; the interaction term ($\text{AI}_i \times \text{Exp}_i$); and four dummy controls: Edu_i , a dummy equal to 1 if developer i has a Bachelor's degree or higher; Size_i ,

a dummy equal to 1 if the size of the i th developer's firm has 1,000+ employees (large); $Industry_i$, a dummy equal to 1 if the i th developer works in the software development industry; and $Country_i$, a dummy equal to 1 if the i th developer lives in a World Bank high-income country (a developed country). ε_i is the stochastic error.

The coefficient of interest is β_3 . In a model with an interaction term, the return to a year of experience on log wages is not just β_2 ; it's β_2 for developers who don't use AI and $\beta_2 + \beta_3$ for developers who do. In other words, a one-year increase in experience raises wages by approximately $\$(\beta_2 \times 100)\%$ for non-AI users and $\$((\beta_2 + \beta_3) \times 100)\%$ for AI users. The sign of β_3 tells me whether AI tools complement or substitute for the human capital that experience represents: if β_3 is positive, AI is amplifying what experienced developers already do, and if it's negative, AI is closing the wage gap between junior and senior developers.

Hypothesized Signs

The following table reports the hypothesized signs for the population parameters.

Coefficient	Variable	Hypothesized Sign	Reasoning
β_1	AI	+	Developers who use AI tools probably earn more because the tools make them more productive, and the kinds of people and firms that adopt AI early tend to be higher-paying anyway.
β_2	Experience	+	More years on the job mean more skill and more on the job knowledge, and the labor market generally pays for that.
β_3	AI $\times$ Experience	-	If AI gives junior developers something close to what senior developers had to learn the hard way, the wage gap between them should shrink. This is the central hypothesis of the paper.
β_4	Education	+	A Bachelor's degree is a credential that employers recognize as higher human capital and tend to pay extra for.
β_5	Size	+	Bigger companies typically pay more than smaller ones for similar work/roles.
β_6	Industry	+	Software companies, where developers are the product, tend to pay developers more than companies where developers are a support function.
β_7	Country	+	Developers in wealthier countries earn substantially more than developers in lower-income countries, and this gap is the single largest source of wage variation in the global data.

Choice of Functional Form

I use the natural log of salary rather than salary alone in dollars for three reasons. First, economists interpret the coefficients in a log-linear model as approximate percentage changes in wages, which is standard in labor economics since the 1974 Mincer wage equation. A \$10,000 raise means something very different to someone earning \$40,000 versus \$400,000. Logs put everyone on the same percentage scale, which matches how people actually think about pay. Second, it's a more meaningful comparison since salaries are right-skewed (a few very high earners pull the mean up), so using the log of salary makes the distribution closer to normal, improving the OLS standard errors. Third, wage variation tends to grow with the wage level, meaning that higher-paid individuals have higher salary ranges than lower-paid ones, and logging compresses that, reducing the severity of heteroskedasticity I'd otherwise run into.

I model experience linearly rather than including a squared term. The Mincer equation typically includes a quadratic term to capture diminishing returns to experience: a worker gains more knowledge and skills in their first year of professional coding, which adds more to wages than in the fifteenth year. By keeping this relationship linear, I am essentially saying that every additional year imposes a constant percentage return, which more certainly oversimplifies what is actually occurring in real life. I made this choice for interpretability: a single coefficient, β_2 , makes the interaction term much easier to understand. I acknowledge this limitation, so I revisit it as a functional-form robustness check and run the appropriate regression in the Results section.

The 2025 Stack Overflow Developer Survey, the dataset I used and will delve more deeply into later, actually asks how often people use AI tools (never / monthly/ weekly/ daily). I simplified this relationship down to a binary variable (yes or no) for one main reason: comparability. It allows me to compare workers who use and don't use AI, building on the previously mentioned literature on AI and productivity, making my results speak directly to theirs.

I also model education, firm size, industry, and country as dummy variables rather than categorical ones. For each of these variables, the survey collected responses to these questions as a short list of multiple-choice options. For example, education included: elementary school, middle school, high school, some college, associate's, bachelor's, master's, PhD, and an 'other' section. I not only collapsed each of these variables into dummies partly for better interpretability, but also because there's a limit to how many variables I can include in the model, and breaking each control into several dummies would be repetitive for this project. Implementing three or four variables only on education or any of the other controls would have come at the expense of other variables with more explanatory power.

Expected Correlations Among Independent Variables

Before running the regression, it is worth flagging which of my variables are likely correlated with each other, since these affect the interpretation of individual coefficients and are something I will check with a VIF test. Larger and software-focused firms are more likely to provide or actively encourage AI tool use, so AI use will very likely positively correlate with company size and the software industry dummy. Microsoft CEO

Satya Nadella reports that roughly 20-30% of the code in Microsoft's repositories is now generated by AI, and Google's Sundar Pichai states that more than 25% of new code at Google is AI-generated (Novet, 2025), and those numbers have only increased further in 2026. Living in a high-income country probably positively correlates with AI use, as the adoption of these tools tends to concentrate in countries with the necessary supporting infrastructure. Taken together, none of these correlations seems severe enough to be a problem on its own. However, they collectively suggest that I should interpret the AI coefficient, β_1 , as the AI premium conditional on a developer's firm, industry, country, and education rather than as the unconditional return from using AI.

Data

Source

The data for this paper comes from the 2025 Stack Overflow Developer Survey, an annual cross-sectional survey of the global developer community administered by Stack Overflow, the largest online discussion platform for programmers. They themselves describe this survey as the “definitive report on the state of software development.” The survey collected over 49,000 responses from 177 countries for 2025. I downloaded the full raw dataset as a CSV file from the official survey results page (<https://survey.stackoverflow.co/>) on March 23, 2026. I did not need any other supplemental data sources, as the survey contains every variable my model requires, including salary, years of professional experience, AI tool use, education, company size, industry, and country.

I chose this dataset for three reasons. First, it is one of the few large-scale sources that asks developers directly about their AI tool use and their wages, which are the exact variables my research question is based on. Second, because it is global and not country-specific, I can capture wage variations across countries that U.S. centered labor surveys rarely include. Third, it is recent enough to reflect the post-AI, post-LLM era of developer workflows, where AI tools have moved from being a developer's toy to being integrated into thousands and possibly millions of people's jobs and routines.

Sample Construction

The raw 2025 survey contained 172 columns of variables covering everything from preferred programming environments to opinions on the Stack Overflow community. After combing through each variable, I dropped 162 columns that were not relevant to my model and kept only the variables tied directly to my regression. I then made several changes to this sample. First, I dropped any observations with missing, zero, or not applicable responses to further clean the data, as those respondents declined to answer. Second, I dropped one outlier who reported 100 years of professional work experience, which isn't a realistic value, as the second-highest response was in the 50s. Third, I dropped respondents who identified as full-time students, freelancers, or retired individuals because these groups aren't the targeted audience for the model to estimate, and the wages for these groups differ meaningfully from those of salaried developers.

Variables and Transformations

The dependent variable is $\ln(\text{Salary}_i)$, the natural log of developer i 's annual salary in U.S. dollars. The Stack Overflow variable 'ConvertedCompYearly' reports salary already converted from local currency to USD, avoiding the need for separate exchange-rate adjustments. I take the natural log for the same reasons discussed in the Model section: log wages have been the standard outcome of labor economics since Mincer (1974), the coefficients are easily interpretable as approximate percentage changes, and it pulls the right-skewed wage distribution closer to a normal distribution.

The first of many independent variables, AI_i is a dummy equal to 1 if the developer reports any use of AI tools in their workflow. The original survey variable 'AISelect' records the frequency of AI use (never/ monthly/ weekly/ and daily), and I keep that version in the dataset under its original name for potential robustness checks. Still, my main regressions use the binary version I constructed from it for reasons discussed in the model section. Experience (Exp_i) is the years of professional work experience, taken directly from the original 'WorkExp' variable. The interaction term ($\text{AI}_i \times \text{Exp}_i$) is the product of the AI dummy and years of experience. For each of the following variables, they were originally categorical; I created dummy versions for reasons previously mentioned in the model section, but kept the original versions in the dataset for potential robustness checks. Education (Edu_i) is 1 if the developer holds a Bachelor's degree or higher. Firm size (Size_i) is 1 if the developer's firm is considered large, with 1,000 or more employees. The industry of the firm (Industry_i) is 1 if the developer works in the software development industry. The country (Country_i) is 1 if the

developer lives in a country classified as high-income by the World Bank Group. I chose the World Bank classification rather than my own because it is external, replicable, and uses a transparent, standardized methodology: the Atlas Method.

Summary Statistics

The following table includes summary statistics for each of my variables after removing irrelevant variables, cleaning the data, and changing the functional form for most variables.

Variable	Minimum	Maximum	Mean	Standard Deviation
Log of Salary	4.62	14.73	11.083	1.133
AI	0	1	0.814	0.389
Experience	1	58	13.209	9.320
AI × Experience	0	58	10.401	9.546
Education	0	1	0.814	0.388
Size	0	1	0.352	0.477
Industry	0	1	0.455	0.498
Country	0	1	0.793	0.404

Results

Main Regression

Using Ordinary Least Squares (OLS), the main regression below estimates the relationship between log developer salary and the seven independent variables in my model. Hypothesis tests are done one-sided at the 5% significance level. With 17,490 degrees of freedom, the critical t-value is around 1.645, meaning any coefficients with t-statistics whose absolute value is greater than 1.645 are considered statistically significant.

$$\ln(\widehat{Salary}_i) = 9.512 - .148AI_i + .02Exp_i + .013(AI_i \times Exp_i) + .184Edu_i + .274Size_i +$$

$$.011Industry_i + 1.318Country_i$$

$$S.E. = (0.032) \quad (0.002) \quad (0.002) \quad (0.018) \quad (0.015)$$

$$(0.014) \quad (0.018)$$

$$t = -4.70 \quad 12.80 \quad 7.06 \quad 10.33 \quad 18.70$$

$$0.80 \quad 74.74$$

$$N = 17,498, \overline{R^2} = 0.3530, F(7, 17,490) = 1,364.98$$

Technical Interpretation of Coefficients

The coefficient on AI (-0.148) indicates that, on average, for a developer with zero years of professional work experience, AI use results in a 14.8% decrease in

salary, holding every other explanatory variable in the model constant. The absolute value of the t-statistic (4.70) is greater than the critical value of 1.645, but because I hypothesized β_1 to be positive, I fail to reject the null hypothesis that $H_0: \beta_1 \leq 0$; AI is statistically insignificant at the 5% level. In other words, the data shows no support for AI substituting for experience for developers at the start of their careers.

The coefficient on Experience (0.020) means that, on average, for a developer who does not use AI, every additional year of professional work experience would result in a 2.0% increase in salary, holding all other explanatory variables constant. Because the t-statistic (12.80) is greater than the critical value of 1.645, I reject the null hypothesis that $\beta_1 \leq 0$; experience is statistically significant at the 5% level.

The coefficient on the AI $\times$ Experience interaction (0.013) implies that, on average, for AI users, every additional year of experience increases their salary by 1.3% than for non-AI users, holding every other variable in the model constant. This suggests that for every additional year of professional work experience, non-AI and AI users' salaries increase by 2.0% and 3.3%, respectively. The t-statistic (7.06) is greater than the critical value of 1.645, but because I hypothesized β_3 to be negative, I fail to reject the null hypothesis that $H_0: \beta_3 \geq 0$; the interaction term is statistically insignificant at the 5% level.

The coefficient on Education (0.184) suggests that, on average, if a developer has obtained a Bachelor's degree or a higher degree equivalent, their salary will increase by 18.4%, holding the other variables in the model constant. Because the

t-statistic (10.33) is greater than the critical value of 1.645, I reject the null hypothesis that $\beta_4 \leq 0$; education is statistically significant at the 5% level.

The coefficient on Size (0.274) is interpreted as, on average, if a developer works for a firm with 1,000+ employees, their salary will be 27.4% higher than it otherwise would, holding the other explanatory variables in the model constant. Because the t-statistic (18.70) is greater than the critical value of 1.645, I reject the null hypothesis that $\beta_5 \leq 0$; size is statistically significant at the 5% level.

The coefficient on Industry (0.011) shows that, on average, if a developer works in the software development industry, their salary will be 1.1% higher than it otherwise would, holding all of the other variables in the model constant. Because the t-statistic (0.80) is not greater than the critical value of 1.645, I fail to reject the null hypothesis that $\beta_6 \leq 0$; industry is statistically insignificant at the 5% level.

The coefficient on Country (1.318) suggests that, on average, if a developer lives in a high-income country, their salary will be 131.8% higher than it otherwise would, holding every other variable in the model constant. Because the t-statistic (74.74) is greater than the critical value of 1.645, I reject the null hypothesis that $\beta_7 \leq 0$; country is statistically significant at the 5% level.

The intercept (9.512) represents the predicted log salary for a non-AI developer with zero years of experience, no Bachelor's degree, working at a small firm in a non-software industry in a non-high-income country.

Qualitative Interpretation

The finding that sticks out to me from this regression is the $AI \times Experience$ interaction term being positive and its t-statistic being greater than the critical value, which directly contradicts my hypothesis that AI tools would substitute for the human capital that experience is proxying for in this model. In this sample, AI tools seem to complement experience rather than substitute for it: every additional year of experience is worth about 65% more in wages for AI users than for non-AI users (3.3% vs. 2.0% per year). It essentially disagrees with the previously mentioned productivity literature on AI and skill, as their findings suggest the opposite of mine.

One thing to note, however, is that β_1 is negative while β_3 is positive, so the total AI effect on wages varies with a developer's experience. This effect is represented as $\frac{\delta \ln(Salary)}{\delta AI} = \beta_1 + \beta_3 \cdot Exp$. If I plug in the estimated coefficients and set this equal to 0, then I get $Exp = \frac{0.148}{0.013} = 11.38$. Essentially, below 11.38 years, the total AI effect is negative, and above 11.38 years, the total AI effect is positive. In other words, below this threshold, AI users earn less than non-AI users; above it, AI users earn more. The effect is negative for early-career developers and positive for late-career developers. The benefit of using AI is basically concentrated among senior developers.

One reason why my findings diverge from the productivity literature is that those studies randomly assign AI access to participants, but in my data, AI use is a choice, and the kinds of people who make that choice may differ in ways the regression can't detect: selection bias. That is to say, maybe the senior developers who use AI are also

more adaptable, curious, or engaged workers who are also more willing to fill out the survey. In that case, my AI coefficient at high experience levels may be partly picking those traits rather than their productivity effect. Conversely, if junior developers who adopt AI are doing so because their skills are weaker than junior developers who don't adopt AI, then the AI effect being negative is reflecting that baseline ability rather than the effect AI itself has on salaries.

A second explanation has more to do with the kinds of tasks where AI benefits developers the most. The productivity literature focuses on small, concrete tasks like resolving customer chats and writing isolated functions, tasks where AI can provide senior-level expertise for junior developers to learn from. Reality, however, is much different. Senior developers spend most of their time on work where AI augments, not replaces them, such as on architectural decisions, code review, mentorship, and debugging old and outdated systems. In these settings, AI multiplies what experience already brings to the table instead of standing in for it. This suggests that my findings may not refute the prior literature, but instead capture real-life work and the nuances that are not controlled for.

The remaining coefficients in the regression mostly land where I'd expect them to. The country coefficient is the single biggest predictor of developer wages, which coincides with Freeman and Oostendorp's (2000) findings that cross-country wage gaps are the dominant source of wage variation across global data. The firm-size coefficient's magnitude is reasonable and statistically significant, which supports Brown and Medoff's (1989) findings on the wage gap across firms of different sizes. The education coefficient's magnitude is also valid and statistically significant, which gives the model

as a whole more credibility. The industry coefficient is the only insignificant variable in my model, making it the prime candidate for the irrelevant variable test discussed below. As I argue below, the variable being insignificant doesn't imply that it's wrong or that it's a flaw; including it in the model, it's just that the country and firm-size dummies are absorbing most of the variation that the software-industry variable would otherwise pick up.

With an $\overline{R^2}$ of 0.353, the model, or the seven independent variables specifically, is explaining about 35% of the variation in log developer salaries. The other 64.7% are factors that the model isn't accounting for: things like individual programming ability, the quality of the firm someone works at, the specific role they hold, and how well they can negotiate contracts and salaries. I explain these issues further in the omitted variable bias section below. The F-statistic of 1,364.98, with a p-value under 0.001, means the model is statistically significant; I reject the null hypothesis that every slope coefficient is equal to zero.

Irrelevant Variables (I.V)

An irrelevant variable is an erroneous independent variable that actually decreases a regression's explanatory power, making it difficult to interpret a model. Out of all the independent variables in the main regression model, industry is the only one whose absolute t-statistic (0.80) isn't greater than the critical value of 1.645. Therefore, to test if this explanatory variable doesn't belong in the model, I dropped the industry variable from the model and reran a regression without it.

$$\ln(\widehat{Salary}_i) = 9.519 - .148AI_i + .02Exp_i + .013(AI_i \times Exp_i) + .184Edu_i + .273Size_i + 1.317Country_i$$

$$S.E. = (0.032) \quad (0.002) \quad (0.002) \quad (0.018) \quad (0.015) \quad (0.018)$$

$$t = -4.70 \quad 12.78 \quad 7.07 \quad 10.32 \quad 18.78 \quad 74.87$$

$$N = 17,498, \quad \overline{R^2} = 0.3531, \quad F(6, 17,491) = 1,592.40$$

First of all, the t-statistic of 0.80 is well below the one-sided critical value of 1.645, meaning that industry essentially adds no statistical explanatory power to the model. Second, in this updated regression with industry being dropped, the $\overline{R^2}$ actually rises slightly from 0.3530 to 0.3531, which is a sign that it was probably an irrelevant variable to begin with since the model held on to its explanatory power even after its removal. Third, removing industry doesn't meaningfully change any of the other coefficients (more than 1 S.E), with the largest change being firm size from 0.2743 to 0.2727 (only 0.0017 or around $\frac{1}{10}$ a standard error). Three of these four criteria point toward labeling industry as an irrelevant variable. However, this last criterion, theory, has made me decide to keep it in my main regression model. Theory suggests that working at software companies, where developers are the product and are generating revenue, tends to pay developers more than companies where developers are a support function and are incurring costs. It's probably the case that the country and firm-size dummies are absorbing most of the variation that the software-industry variable would otherwise pick up. Therefore, keeping industry in the primary model costs nothing essentially in terms of explanatory power, while the cost of omitting a

theoretically relevant variable is possibly more devastating, since omitted variable bias can affect other coefficients of interest I really care about: AI × Experience.

Multi-Collinearity

Perfect multicollinearity occurs when two or more independent variables are perfect linear functions of each other; my model doesn't run into this, supported by the fact that Stata didn't drop a variable. Because I am working with cross-sectional data, I will test if any imperfect multicollinearity exists using the correlation coefficients and the Variance Inflation Factor (VIF) on the primary regression model. If an absolute correlation coefficient is greater than 0.8 or if a VIF is greater than 5, then these tests indicate that there may be a very strong but not perfect relationship between two or more explanatory variables, inflating S.E's and making the estimated coefficients sensitive to changes between samples.

The highest correlation coefficient between any of the variables is between Experience and the AI × Experience interaction term at 0.696, which isn't above the 0.8 threshold. The next-highest correlation is between AI and AI × Experience at 0.521, again with the interaction term. Other than those two, most of the other correlation coefficients fall below 0.15, suggesting that the model is free of imperfect multicollinearity. Below, I present the VIFs from my main regression.

Variable	AI	Exp	AI × Exp	Edu	Size	Industry	Country	Mean VIF
VIF	3.17	4.45	6.05	1.01	1.03	1.04	1.07	2.55

As we can see, four control variables, Country, Industry, Size, and Education, all have very low VIFs, not even close to 5; there is no meaningful relationship among these control variables. However, AI, Experience, and the AI $\times$ Experience interaction term all have relatively higher VIF values compared to the control variables, with the interaction term actually being greater than the threshold ($6.05 > 5$).

Although the VIF test suggests that the interaction term violates imperfect multicollinearity, in reality, by definition, I expect this inflated number. Since the interaction term is the product of AI and Experience, it naturally correlates with both AI and Experience. The interaction term constructs a high VIF number, and I should not treat it as a real problem. Therefore, there is no need to correct for imperfect multicollinearity; the standard errors are narrow, the coefficients on AI, Exp, and AI $\times$ Exp are statistically significant, and the 17,498 observations provide a lot of statistical power.

Heteroskedasticity

Heteroskedasticity, both pure and impure, occurs when the variance of the error term is not constant across observations; the standard errors become biased, which also biases the resulting t-statistics and confidence intervals. To check whether my main regression has any pure heteroskedasticity, I run the Breusch-Pagan test, with the null hypothesis defined as the variance being constant. Because the resulting p-value is less than 0.05, I will reject the null hypothesis, indicating that the main regression has pure heteroskedasticity.

I already foresaw this outcome because, as I have previously mentioned, wage variance tends to grow with wage level itself, with experience, and with country and firm characteristics. I logged the dependent variable in the main regression to begin with to compress this variance growth, but it still did not fully eliminate the problem.

To correct for heteroskedasticity, I re-estimated the regression using robust standard errors, which don't change the coefficient estimates but correct the standard errors and t-statistics. I report this new regression below.

$$\ln(\widehat{Salary}_i) = 9.512 - .148AI_i + .02Exp_i + .013(AI_i \times Exp_i) + .184Edu_i + .274Size_i + .011Industry_i + 1.318Country_i$$

<i>Robust S.E</i>	=	(0.030)	(0.001)	(0.002)	(0.017)	(0.015)
			(0.014)	(0.022)		
<i>t</i>	=	- 4.96	14.60	7.69	10.53	18.73
			0.80	58.67		

$$N = 17,498, \overline{R^2} = 0.3530, F(7, 17,490) = 864.06$$

After correcting for the standard errors and t-statistics, all coefficient estimates in the main regression under OLS are essentially the same; every statistically significant coefficient stays significant with robust standard errors, and the only statistically insignificant coefficient, industry, stays insignificant with a seemingly identical t-statistic of 0.80.

For the AI, Experience, AI $\times$ Experience, Education, and Size variables, the robust standard errors came in slightly smaller than the OLS standard errors, meaning OLS overestimated them by a small margin, but the correction narrowed down these estimates to be more accurate. However, for the Country variable, its robust standard error actually came in meaningfully bigger than its OLS version, rising from 0.018 to 0.022 (a 28% increase), with its resulting t-statistic falling from 74.74 to 58.67. This suggests that the heteroskedasticity in the residuals mostly lies within this variable; the variance of log wages is a lot higher in high-income countries than in lower-income countries. This fits with the wider spread of developer salaries in countries where the upper end of the wage distribution stretches into the hundreds of thousands and even into the millions, so I would end up preferring this model over the main regression model.

Functional Form (F. F)

As previously stated, the foundational Mincer (1974) wage equation in modern labor economics includes both Experience and Experience² as independent variables to capture the diminishing returns to experience. The idea is that a worker's first year on the job adds more to their wages than their fifteenth, and so modeling this relationship linearly fails to capture the underlying concavity. My main regression sticks with a linear specification for simplicity and interpretability; however, by doing so, I may be incorrectly specifying this relationship and biasing the AI $\times$ Experience interaction coefficient. To test this, I add Exp_i² to the main regression on its own while leaving the interaction term as it is. The model is reported below.

$$\ln(\widehat{Salary}_i) = 9.148 - .059AI_i + .083Exp_i - .0016Exp_i^2 + .0054(AI_i \times Exp_i) + .163Edu_i + .272Size_i + .012Industry_i + 1.286Country_i$$

$$S.E = \begin{matrix} (0.031) & (.0029) & (.000063) & (0.0018) & (0.018) \\ & (0.014) & (0.014) & (0.017) & \end{matrix}$$

$$t = \begin{matrix} -1.89 & 28.78 & -25.79 & 3.07 & 9.30 \\ & 18.88 & 0.86 & 74.16 & \end{matrix}$$

$$N = 17,498, \overline{R^2} = 0.3767, F(8, 17,489) = 1,322.81$$

As we can see, this regression model fits the data a lot better than the main regression. $\overline{R^2}$ rises from 0.3530 to 0.3767 (a 7% increase), and the coefficient on experience squared also inherits the negative sign that almost every empirical study using the Mincer equation predicts; experience is increasing wages at a decreasing rate. As a result, these findings, in tandem with the prior literature, support the idea that the Mincer form of specifying experience is the more appropriate functional form for this data than the linear specification I assumed in my primary regression model.

To illustrate, the AI coefficient decreases from -0.148 to -0.059, with its t-statistic dropping from -4.70 to -1.89, which is still above the critical value of 1.645. The AI × Experience interaction's coefficient also decreases from 0.0125 to 0.0054, with its t-statistic dropping from 7.06 to 3.07. This pattern happens when experience enters the regression with the wrong functional form, since the interaction term ends up absorbing

some of the nonlinearity that experience squared should be accounting for, as shown with the inflated coefficient. By correctly controlling for this nonlinearity with experience squared, the model produces more conservative coefficients while preserving the qualitative finding that every additional year of experience is worth more in wages for AI users than for non-AI users.

The remaining coefficients stay mostly the same in this updated regression. Education decreases slightly from 0.184 to 0.163; Size, Industry, and Country all change by less than two percent; Industry remains insignificant, indicating that the linearly specified experience in the main regression didn't really affect these control variables. Therefore, I would end up preferring this model over the main regression model: it fits the data better and lines up with standard labor economic practice, the qualitative interpretation stays the same, and the overall central finding that AI complements rather than substitutes for the human capital that experience represents holds, with only the details needing to be updated.

Omitted Variable Bias (OVB)

Omitted variable bias occurs when a variable that affects the dependent variable is not included in the regression but happens to be correlated with one or more of the included variables. The amount of expected bias on a coefficient $\hat{\beta}_1$ from an omitted variable is given by

$$\text{Bias} = \beta_{\text{omitted}} \times \alpha_1$$

where β_{omitted} is the omitted variable's impact on the dependent variable, and α_1 is the included variable's impact on the omitted variable. The expected bias is positive when both terms have the same sign and negative when they are different. By definition, you can't quantify omitted variables as they are unobserved (perhaps you don't have data on the omitted variable); therefore, one must reason which variables are likely missing, hypothesize what their expected bias is on each coefficient, and deduce if estimates can be read as causal vs. correlational. I will focus on my AI ($\hat{\beta}_1$) and interaction ($\hat{\beta}_3$) coefficients below because these are two of the most significant coefficients of interest for this model.

To start, one of these omitted variables most likely includes a developer's abilities/skills. More capable and talented developers probably earn higher salaries (β_{omitted} is +), and they may also be early adopters of new tools and systems to get ahead of other developers (α_1 is +). Therefore, the expected bias on $\hat{\beta}_1$ is positive (+ $\times$ + = +). In other words, my AI coefficient is biased upwards; it is capturing some of the effects on wages attached to higher-ability developers. This positive expected bias would also extend to the interaction term, because, as was previously mentioned, the interaction term inherits the characteristics of its factors, like AI.

Additionally, another likely source of omitted variable bias is firm quality and AI policy. Better-resourced firms pay more (β_{omitted} is +) and are also more willing to provide or encourage AI tools amongst their developers (α_1 is +); Microsoft and Google shared that 20-30% of their code is AI-generated (Novet, 2025). The product of two positives is positive, meaning the expected bias on $\hat{\beta}_1$ is positive. The size dummy

variable will capture some of this effect, as larger firms tend to be better-resourced, but this isn't to say every large firm is better resourced or that the quality amongst large firms is the same. Whatever is left, the variation in firm quality that size doesn't account for probably gets reflected with a positively biased AI coefficient.

Although so far these omitted variables predict a positively biased AI coefficient, it does not explain why the coefficient is negative in the main model. It may be primarily due to functional form, as after correcting for experience's functional form, $\hat{\beta}_1$ decreased from -0.148 to -0.059 and almost pushed its absolute t-statistic (1.89) to be lower than the critical value of 1.645. However, an omitted variable that may be negatively biasing AI downwards would be job type/role, such as being an individual contributor vs. a manager. Product managers and engineering managers typically earn more than individual contributors at the same experience level because managers have to strategize what to prioritize and what needs to be accomplished, and they have greater responsibility over individuals and over cost/budget ($\beta_{omitted}$ is +). Managers probably spend less of their day writing and debugging code than individual team members do, so they are less likely to use AI (α_1 is -). Therefore, the expected bias on $\hat{\beta}_1$ is negative; the product of a positive and a negative is negative.

Conclusion

Motivated by the seemingly exponential development and integration of AI tools into developer workflows, this paper answers: Do AI tools affect wages differently depending on developer experience? Using a cross-sectional sample of 17,498 developers from the 2025 Stack Overflow Annual Developer Survey, I used a Log-linear regression model with log salary as the dependent variable and independent variables like an AI dummy, years of experience, an AI $\times$ Experience interaction term, and binary controls for education, firm size, industry, and country.

Building from the empirical productivity literature on AI (Brynjolfsson, Li, and Raymond 2025; Peng et al. 2023; Cui et al. 2024), I expected the interaction coefficient β_3 to be negative, indicating that AI substitutes for the human capital that experience represents and narrows the wage gap between junior and senior developers. Instead, the coefficient came out positive and statistically significant, contradicting my initial hypothesis. In the main regression, every additional year of experience pays 2.0% more for non-AI vs. 3.3% for AI users, which also holds after correcting for heteroskedasticity and functional form in experience. In terms of the overall AI effect, below a threshold of 11.38 years of experience, AI users earn less than non-AI users, and above that, AI users earn more; the benefit of using AI is essentially concentrated among senior developers.

The remaining binary controls mostly line up with what prior literature predicts. Living in a high-income country is the single biggest predictor of developer wages, which is supported by Freeman and Oostendorp's (2000) findings that cross-country

wage gaps dominate global wage variation. Working at a large firm (1000+ employees) on average results in a 27% higher salary, supporting Brown and Medoff's (1989) findings on the wage gap across firms of different sizes. Holding a Bachelor's degree typically results in a 18% higher salary, which is reasonable. Industry is the only insignificant variable in my model, which can be a result of the Country and Size dummy variables absorbing most of the variation that Industry would otherwise pick up.

The cleanest contribution this paper showcases is that the documented AI productivity effect apparent in the prior literature does not appear to translate to actual developer salaries. In those studies, AI closes gaps. In the salary data, it widens them. 2 possible explanations are evident. First, those studies randomly assign AI access to participants, which strips away any unobserved traits between AI adopters and non-adopters, but in my data, AI use is a choice; the senior developers who picked up AI are probably also the adaptable and curious ones in ways junior developers may not be. Second, those studies measure productivity narrowly, focusing on small and concrete tasks where senior-level expertise is relatively intuitive to pass down. Real senior developer tasks, however, are different from those in the controlled studies; more time is dedicated to tasks like architectural decisions, code review, and mentorship, areas where AI amplifies what experience already brings to the table instead of substituting for it.

One direction future research could follow would include controls for job roles (front-end vs. back-end vs. machine learning vs. management), which could meaningfully change the AI and the resulting interaction coefficient, as there is no proxy that currently measures this in the main model. As I mentioned in the OVB section,

managers earn more than individual contributors but use AI less, which biases the AI coefficient downward. Adding role controls would remove this bias, moving the AI coefficient closer to its true value and changing how AI users compare to non-users across the experience distribution. As AI tools make their way deeper into developer workflows, what's at stake goes beyond individual career decisions; it shapes how economists think about AI and the role it has in the economy.

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Data Appendix

All data for this paper comes from the 2025 Stack Overflow Annual Developer Survey. I also used the World Bank Group's country income classification when transforming the country variable. All data was accessed and downloaded on March 23, 2026.

Primary Data Source

The 2025 Stack Overflow Developer Survey

To replicate the dataset:

1. Navigate to <https://survey.stackoverflow.co/>.
2. Click into the "2025" section of the results page.
3. Scroll and download the full raw dataset in CSV format. The file contains all raw survey responses for 2025.
4. Open the CSV file in Stata, Excel, or any program of your choice.

Sample Construction

Starting from the raw 2025 survey CSV file (172 columns and 49,000+ rows), these are the following changes I made:

1. Dropped 162 irrelevant columns, only kept: ConvertedCompYearly, WorkExp, AISelect, EdLevel, OrgSize, Industry, Country, Employment, and DevType.
2. Dropped observations who did not identify as employed.
3. Dropped observations with missing, zero, other, or "not applicable" responses.
4. Dropped outliers who reported 0 or 100 years of professional work experience.

5. Dropped observations who identified as full-time students, freelancers, or retired individuals.

The final sample contains 17,498 observations.

The full data-cleaning pipeline, including the Python script used for column removal and row filtering, the exact sequence of commands run, and a step-by-step reproduction guide, can be found in `data_processing/README.md`, included with the submission.